Midterm Examination
ECO 101: Principles of Microeconomics
Department of Economics
Shiv Nadar University
Monsoon 2017.

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This is a one and a half hour long closed book, closed notes in class examination. Answer all the questions, and be clear and precise in your answers. Any act of *academic dishonesty* will be taken very seriously. *All the best.*

Question 1: Incidence of Commodity Taxation (20 points)

Consider the following model for the toothbrush market in Delhi. Suppose the aggregate demand for brushes in Delhi is given by $Q_D = 900 - P/2$ where P denotes the price and Q denotes the quantity of brushes. The aggregate supply for brushes in Delhi is given by $Q_S = P/4$.

(a.) Compute the toothbrush market equilibrium. What are the equilibrium price and quantity? (4 points)

(b.) Now suppose a tax of $t = 60$ is imposed on each brush that is purchased. Write down the changed post-tax demand and supply equations and compute the brush market equilibrium with the tax. What are the equilibrium price that the sellers receive, and what is the post-tax equilibrium quantity transacted? (6 points)

(c.) What is the price paid by the buyers and therefore, what is the incidence of the tax? Explain the intuition behind the key factors that determine the incidence. (5 points)

(d.) Compute and graphically depict dead-weight loss due to the tax. (5 points)

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Question 2: Consumer Choice and Comparative Statics (30 points)

Consider three siblings, Ryan, Liz and Emma, who all get INR 100 per week to spend on the goodies they like to consume. The only two consumables they purchase out of that money is Chocolate (X) and Ice-cream (Y)[1]. The price of an unit of Chocolate (X) is 2 whereas that of an unit of Ice-cream (Y) is 4. $2X + 4Y = 100$

(i) Ryan's preference over the two goods are given by the following utility function: $u^r = \min\{x, y\}$. Draw the family of indifference curves for Ryan, and precisely explain the characteristics of his preferences that gives the shape of the IC's you obtain. (3 points)

(ii) Liz's preference, on the other hand, is represented by the following utility function: $u^l = \max\{x, y\}$. Draw the family of indifference curves for Liz, and precisely explain the characteristics of his preferences that gives the shape of the IC's you obtain. (4 points)

[1] With a slight abuse of notation, we denote by the capital letters the good and by small letters the amount of that good consumed

(iii) Point out the most crucial difference between the characteristics of their preferences and explain, economically, how that would make a difference in the bundles they choose to consume. (3 points)

However, in this question, we are interested in studying the choices made by Emma, the youngest of them all. Since she is too young to make her own choice, she follows what both of her brothers individually do, and therefore, her preferences over these two goods are given by: $u^c = \min\{x, y\} + \max\{x, y\}$

(iv) Draw the indifference curves for Emma and explain the shape of the IC's in terms of the characteristics of her preferences. (5 points)

(v) With the initial prices, $(p_x = 2, p_y = 4)$, how many units of Chocolate and Ice-cream will Emma consume? Briefly explain her choice. (5 points)

(vi) Suppose the price of X rises to 5 per unit, keeping that of Y same as before at 4 and her pocket money unchanged at 100. Find the new equilibrium consumption by Emma. (5 points)

(vii) Decompose this price effect into income and substitution effects and briefly explain your findings. (5 points)

Date: September 27, 2017.
Instructor: *Subhra K. Bhattacharya.*

$x + y = 1$

$2 \times \frac{100}{6} + 4 \times \frac{100}{6}$

$x + y = 1$

$y = 1 - x$

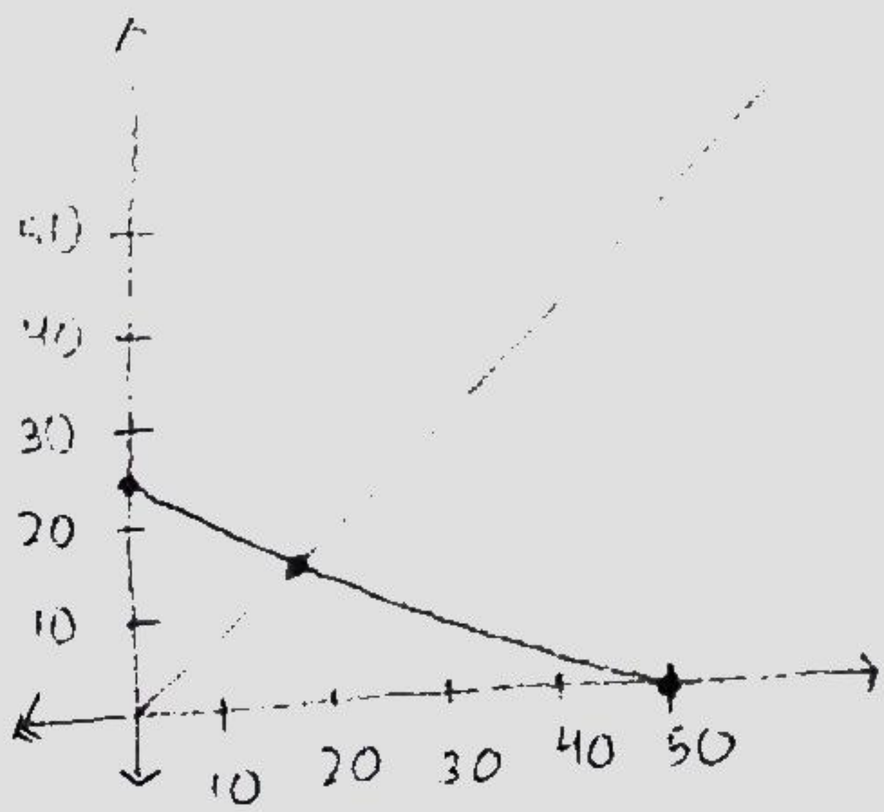
$2x + 4y = 100$

$6y = 100$

$y = \frac{100}{6}$

$\frac{x}{y} = 1$

$y = \frac{100}{6}$



$5x + 4y$

$\frac{100}{6}$ 3

$2xy$

$x + y =$

$\frac{1+y}{1-x}$